IE 643 COURSE PROJECT · PREP PRESENTATION

Probing the Latent Space of Video Diffusion U-Nets

Disentangling content vs. motion

Team Claudn't

Roll numbers — [FILL IN, one per team member]

This work is carried out as part of the IE 643 course project.

21 September 2026

OUTLINE

What we cover, and in what order

01 Why content and motion, and what we mean by each

02 How a video diffusion U-Net is put together

03 Activation patching and cross-attention as tools

04 Where motion has already been found by others

05 Model choice, and a controlled probing set

06 Four pilot experiments we have already run

07 The four deliverables, and how each feeds the next

08 Metrics, baselines, and what would count as success

09 Roadmap, current status, risks, references

THE PROBLEM

Two things a video model has to get right at once

A video diffusion U-Net alternates between two kinds of layer. Spatial layers work inside a single frame and settle what the scene looks like. Temporal layers work across frames and settle how it moves.

If we can say which layers and which denoising steps hold which, we can change one and leave the other alone. That edit is the test.

OUR RUNNING EXAMPLE

"A golden retriever running across a beach." Keep the gait, turn the dog into a cat. Or keep the dog and slow the gait to a walk. Either edit should leave the other axis untouched.

Content

Object identity, colour, texture, scene layout. What is depicted.

Dog identity, fur colour, the beach and sky behind it.

Motion

Temporal dynamics across frames. How things move, deform and travel, whatever is depicted.

The running gait, and the path it traces across the frame.

How a video diffusion U-Net is put together

The architecture factors the problem. Spatial blocks are inherited from image diffusion and see one frame at a time. Temporal blocks are the only path along which information crosses frames — and the pattern repeats at every resolution and every timestep.



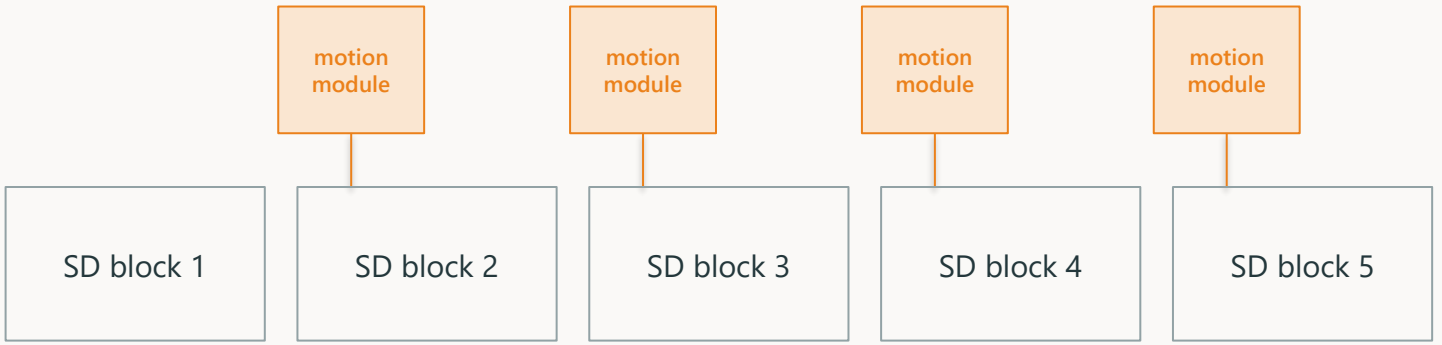
encoder → bottleneck → decoder, one resolution level shown

Ho et al., Video Diffusion Models (NeurIPS 2022) · Blattmann et al., Align your Latents (CVPR 2023) · Blattmann et al., Stable Video Diffusion (2023)

What this means for us. The split is already there structurally. Our contribution is to show it also holds causally, which is a different and much stronger claim. A layer can be labelled temporal and still carry appearance, and we should expect some of that.

Motion modules: a split that is partly given

AnimateDiff takes a pretrained image model, freezes it, and threads a separately trained motion module between its blocks. Content and motion live in different parameter sets by construction, which is the handle we grab first.



frozen image-diffusion backbone — the content path

SIZES, CORRECTED

SD1.5 UNet ≈ 860M, frozen

Motion module ≈ 417M

`mm_sd_v15_v2, 1.6 GB on disk`

Our earlier draft said 1.7B. That was the file size, not a parameter count.

Guo et al., AnimateDiff (ICLR 2024) · Zhao et al., MotionDirector (ECCV 2024) · Wu et al., Tune-A-Video (ICCV 2023)

The catch we have to be honest about: the backbone was frozen, so its layers never saw motion during training. Some of the separation is an artefact of that recipe, which is why a jointly-trained model stays on the list.

Activation patching: turning a guess into a cause

Run the model on clip A and cache what one layer produced at one timestep. Run it again on clip B, but splice A's cached activation in at that same coordinate. If B's output drifts towards A along one axis and not the other, that coordinate carries that axis.



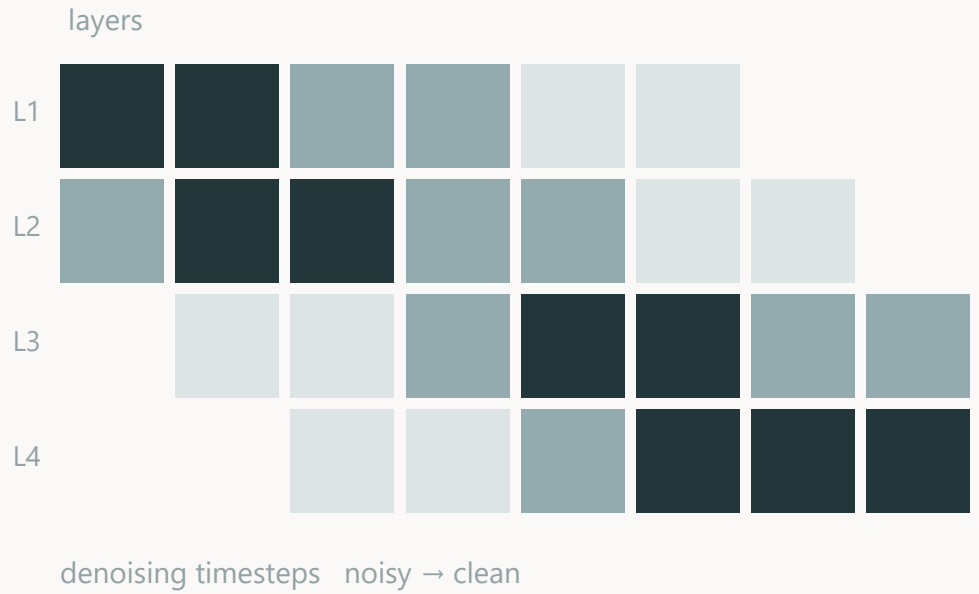
Then ask: does the output now look like A, move like A, both, or neither?

Meng et al., ROME (NeurIPS 2022) · Wang et al., IOI circuit (ICLR 2023) · Zhang & Nanda, Best Practices of Activation Patching (ICLR 2024)

All three come from language-model interpretability, and the logic moves across unchanged. We are reading Zhang and Nanda before writing any patching code — their warnings about metric choice and about noising versus resampling corruption describe mistakes we would otherwise make.

Cross-attention as a diagnostic lens

Cross-attention maps show which regions and which frames a text token is read into, at every layer and every denoising step. A cheap way to see what a layer is looking at before we commit to intervening on it.



Illustrative. Real maps come out of the harness in Stage 2.

Hertz et al., Prompt-to-Prompt (ICLR 2023)
Tang et al., DAAM (ACL 2023)

Both give working recipes for pulling attention out of an image U-Net. Our change is small but it matters: we keep the frame axis instead of collapsing it, so a token's map becomes a short movie rather than a still heat map.

Where motion has already been found

Reading this group changed our plan. Three results tell us roughly where to look before we start, which turns a blind sweep into a set of predictions we can be wrong about.

LAYER

Mid-resolution, not the extremes

Xiao et al. build a motion feature from U-Net activations and find the cleanest signal just after the first up-block. Deeper and shallower blocks both do worse.

Xiao et al., NeurIPS 2024

TIMESTEP

Motion early, appearance late

Widely assumed, rarely tested. Baherwani et al. measured it and report two regimes with a reasonably stable boundary between them.

Baherwani et al., arXiv 2025

SIGNAL

Attention carries the motion

MotionClone finds the dominant components of temporal attention drive motion; Yatim et al. get a motion descriptor by averaging space-time features over space.

Ling et al., ICLR 2025 · Yatim et al., CVPR 2024

And the older framing. MoCoGAN (2018) split content from motion by hand in the generator's latent code, and CMD (ICLR 2024) does the diffusion-era version of the same idea. Both build the split in. We are asking whether a model that was never factored this way learns a version of it anyway, which is the harder question.

SETUP

Model choice, and the data we probe with

AnimateDiff is our primary model. The spatial and temporal paths are already separate modules in the code, so Deliverable 2 starts from a hypothesis instead of a blank sheet.

MODEL	SIZE	WHY IT IS ON THE LIST	WHY IT IS NOT PRIMARY
AnimateDiff PRIMARY	417M motion module + ~860M SD1.5 UNet	Explicit spatial/temporal split; one 12–16 GB GPU; mature diffusers support	Motion module sits on a frozen image model, so the split is partly given
ModelScope T2V	1.7B	Spatial and temporal trained jointly, end to end. The harder test case	Thinner tooling for attention extraction than SD-derived pipelines
Stable Video Diffusion	~1.5B UNet	Best output quality of the three, well documented	Image-conditioned only, so matched text-prompt pairs are not possible

THE PROBING SET WE GENERATE OURSELVES

About 120 clips on a prompt grid of the form [subject] [verb] [scene]. Hold two axes, vary the third: a content pair changes only the subject, a motion pair only the verb.

Subject and scene vocabulary from MSR-VTT and WebVid-10M captions; motion verbs from UCF101 and Kinetics-400 labels. We train on none of these — they are a source of vocabulary and structure only.

WORK SO FAR

Four pilot experiments, 2–10 September

Small and deliberately cheap. The point was to find out whether the project is tractable, not to produce results.

	QUESTION WE WERE ASKING	WHAT WE RAN	WHAT WE SAW
E1	**Can we generate video at all on our hardware?**	AnimateDiff v1.5 through diffusers. 16 frames, 512×512, 25 DDIM steps.	Runs. [__ s/clip, __ GB VRAM]. This sets the budget for every later sweep.
E2	**Does a fixed seed hold the scene still when we change one word?**	Same noise tensor; prompt varied on one axis at a time (retriever→cat, running→walking).	Largely yes. Enough for paired clips to work as a controlled design.
E3	**Can we get activations out, and what does storing them cost?**	Forward hooks on up_blocks[i].motion_modules[j] and the attn2 modules; dumped shapes and sizes.	Easy to hook. Storage, not compute, is the binding constraint. [__ GB/clip]
E4	**Do the temporal modules causally carry motion?**	Scaled the motion-module residual towards zero, everything else identical.	The clip goes near-static while each frame stays a plausible image.

Three decisions came out of this. AnimateDiff is primary (E1, E4); every probing clip shares a fixed seed and noise tensor (E2); the harness stores summaries by default, not raw tensors (E3).

PART TWO

Our approach

Four deliverables. Each one is only useful because of the one before it.

01	**Setup and probing tools**	Configure the model, hook every spatial and temporal block, dump attention per timestep.
02	**Content / motion analysis**	Score every (layer, timestep) for whether it tracks what is depicted or how it moves.
03	**Patching framework**	Transplant the coordinates that scored highest, from one clip's generation into another's.
04	**Disentanglement demonstration**	Keep the motion and change the content, then the reverse, and measure both directions.

DELIVERABLE 1

Setup and the extraction harness

01 Hook both paths by name

diffusers builds a UNetMotionModel from a UNet2DConditionModel plus a MotionAdapter. Temporal path: the motion_modules inside each down_blocks[i], up_blocks[i] and the mid block. Spatial path: the transformer_blocks, with attn1 and attn2 attention.

02 Index everything by one coordinate

A triple of (module path, denoising timestep, clip id). Every later stage queries by that triple, so it is worth getting right once.

03 Store summaries, not raw tensors

E3 ruled out a full dump. We keep channel statistics, frame-mean-removed norms and downsampled attention maps, with full tensors only where we flag them.

04 Pin down determinism

One seed policy, enforced in the generation script. Each clip records its seed, sampler, step count, resolution and model revision next to its activations.

Status. Hooks and generation prototyped in E1 and E3. The production harness is Stage 2 work, 28 September to 4 October.

DELIVERABLE 1

Building a probing set that can answer the question

The probe measures how activations differ between two clips. If anything other than the axis we care about also differs, the measurement is worthless. So the set is built around one rule: change exactly one thing.

THE PROMPT GRID

[subject] [verb] [scene]

CONTENT PAIR — VARY THE SUBJECT

`a golden retriever running on a beach`
`a horse running on a beach`

MOTION PAIR — VARY THE VERB

`a golden retriever running on a beach`
`a golden retriever walking on a beach`

Same seed, same noise tensor, across a pair.

Without this, the difference between two clips is mostly sampling noise and the probe measures nothing. E2 is what told us this is viable.

Roughly 120 clips: 15 subjects × 4 verbs × 2 scenes.

Generated once, reused for probing, patching and evaluation. Generation time is the bottleneck, not data.

Vocabulary borrowed, not invented.

Subjects and scenes from MSR-VTT and WebVid-10M captions; verbs from UCF101 and Kinetics-400 labels.

Pairs are labelled by construction, which is what makes a quantitative probing metric possible at all.

DELIVERABLE 2

Three predictions we can be wrong about

Sweeping blind produces a description. Going in with predictions produces a result. Each of these comes from the background reading, and each is cheap to falsify.

H1	**Layer axis**	Motion selectivity peaks in the temporal modules at medium resolution, around up_blocks[1], rather than at the highest-resolution decoder layers or at the bottleneck.	*Follows Xiao et al. Falsified if the peak sits at an extreme, or if it is flat.*
H2	**Timestep axis**	Motion is settled in roughly the first third of the denoising trajectory and appearance in the last. Content selectivity should run the opposite way.	*Follows Baherwani et al. If our boundary sits elsewhere, that is a finding, not a bug.*
H3	**The confound**	Raw activations look content-selective everywhere. A clean motion signal only appears after subtracting the per-frame mean, which removes the static component.	*The one that matters most. Skip it and we would wrongly conclude motion is nowhere.*

H3 is the expensive one. If we had not read Xiao et al. we would have spent the first week of the sweep concluding that every layer encodes appearance.

DELIVERABLE 2

One number per layer, per timestep

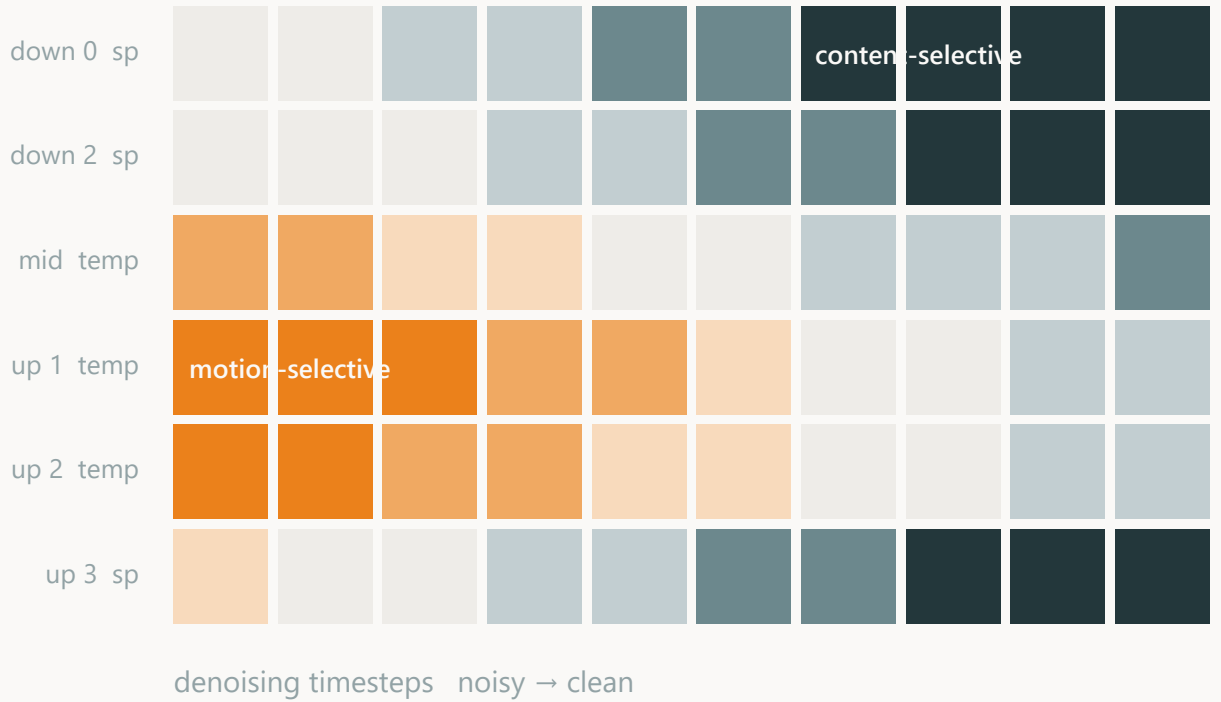
For each coordinate, take a content pair and a motion pair that share a seed. Let d_c be how far the activations move under the content swap, and d_m how far under the motion swap.

$$S(L, t) = (d_c - d_m) / (d_c + d_m)$$

The sign is what makes this useful. Positive means the coordinate tracks what is depicted, negative means it tracks how things move, near zero means both or neither. A raw divergence score cannot tell those apart.

AND A SECOND, INDEPENDENT VIEW

A small linear probe per coordinate says what information is present. Patching says what the model actually uses. Where they disagree is likely the most interesting thing we find.



The pattern H1–H3 predict, drawn by hand. Stage 4 replaces it with measured values, and the interesting outcome is the one where it does not match.

DELIVERABLE 3

The patching and intervention framework

A patch swaps the activation at one coordinate during clip B's generation for the one cached from clip A. Both runs must start from the same noise tensor, or the comparison means nothing.

FOUR GRANULARITIES, COARSE TO FINE

1. **Whole tensor**
   Blunt, probably breaks coherence, but it bounds the effect size.
2. **Channel subset**
   Only the most motion-selective channels. We expect this is where it starts working.
3. **Attention weights only**
   Swap the temporal attention map rather than the features, after MotionClone.
4. **Scaled injection, $\alpha \in [0,1]$**
   Blend instead of replace. Sweeping α gives a dose-response curve.

CONTROLS WE RUN FIRST

Null patch

Patch a clip with its own activation. The output must come back identical. If it does not, the harness has a bug and nothing downstream is trustworthy.

Random coordinate

Patch somewhere with low selectivity. If that works as well as our chosen coordinate, we have not localised anything.

Symmetry

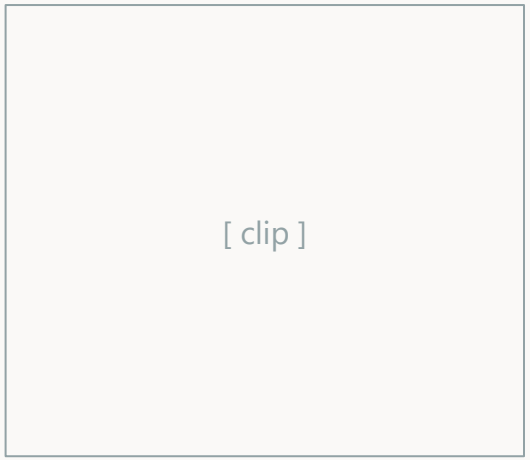
A into B, then B into A. The two effects should be near inverses. Asymmetry means something is leaking.

Status. Design is settled. Implementation is Stage 5, 19 to 25 October.

DELIVERABLE 4

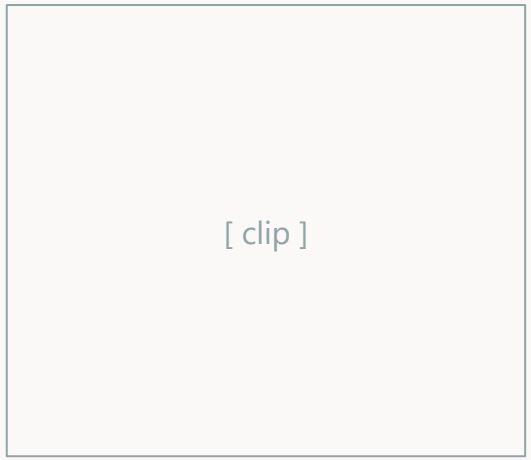
What the demonstration looks like

Two edits per test case, each with an unambiguous target. Reported as a grid, and scored on the next slide.



Original A

golden retriever, running



Original B

horse, walking



Content of B + motion of A

horse, running



Content of A + motion of B

golden retriever, walking

Both directions have to work. A method that transfers motion but not content has not demonstrated disentanglement, it has found one useful lever. Placeholders get replaced with real frames in Stage 6, 26 October to 1 November.

EVALUATION

Metrics, and what we compare them against

Content fidelity

CLIP similarity to the intended content prompt, plus CLIP or DINO image-to-image similarity against the reference frames. The image-to-image version is the honest one.

Motion fidelity

Trajectory-based, after Yatim et al.: extract point tracks with CoTracker and compare displacement vectors over time. RAFT flow correlation as a cheaper second opinion.

Did we wreck the clip?

Two or three VBench dimensions — subject consistency, motion smoothness, temporal flickering. Cheap to run, and it answers the obvious objection before it is raised.

THE HEADLINE NUMBER

$$D = \tfrac{1}{2}[\, s_c(\text{out},B) + s_m(\text{out},A) \,] - \tfrac{1}{2}[\, s_c(\text{out},A) + s_m(\text{out},B) \,]$$

For an output meant to take B's content and A's motion. Large and positive means we moved the axis we aimed at and left the other alone. Reported for both directions.

BASELINES — WITHOUT THESE THE NUMBER MEANS NOTHING

Prompt-only

Same seed, new prompt, no patching. If we cannot beat this, we have built an expensive way to do nothing. Mandatory.

Training-free

MotionClone or the MOFT controller. Public code, comparable in spirit to what we are doing.

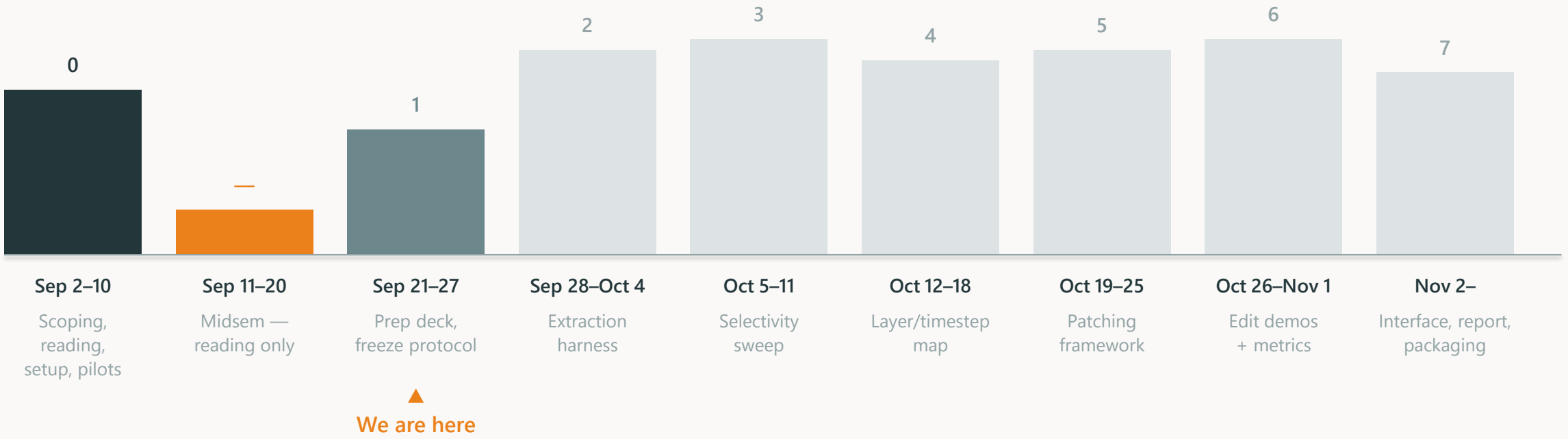
Trained

MotionDirector, if time allows. A ceiling we do not expect to reach, and we will say so.

ROADMAP

2 September to the intensive assessment

Bar height is planned effort. The dip is the midsem window, where we deliberately scheduled reading and nothing else.



Where the slack is. Stage 2 is likeliest to overrun. If it does, the time comes out of Stage 5, not Stage 3 — a rushed patching implementation is recoverable, a rushed probing sweep is not.

STATUS

Where we actually are today

Background reading	**Done**	Groups 1–5. The recent motion-localisation papers changed the plan.
Model selection	**Done**	AnimateDiff primary, ModelScope T2V held back as a robustness check.
Environment, baseline generation	**Done**	E1 — AnimateDiff runs end to end in diffusers on our GPU.
Pilot experiments	**Done**	E2–E4. Seed control verified, hooks prototyped, motion module ablated.
Extraction harness (production)	Stage 2	28 Sep – 4 Oct. Prototyped in E3, not yet built properly.
Probing sweep, selectivity map	Stages 3–4	5–18 Oct. The intellectual core of the project.
Patching framework	Stage 5	19–25 Oct. Designed, not implemented.
Edit demos, metrics, interface	Stages 6–7	26 Oct onwards.

Numbers in the pilot table are still blank in places. Those get filled from our own logs before the recording, not estimated.

RISKS

What could go wrong, and what we do about it

Compute limits the layer × timestep sweep

→ Start coarse — every other layer, every fourth timestep — and refine only around what looks promising. E1's timing numbers tell us what we can afford.

Patched outputs come out incoherent rather than wrong

→ Run the null patch and the random-coordinate control before believing anything. Prefer scaled injection over hard replacement, and report the whole α curve.

AnimateDiff's split is too easy to be a finding

→ Fair criticism, and we say so up front. The answer is ModelScope T2V, where spatial and temporal were trained together. If the same structure shows up there, it generalises.

Everything looks content-selective and we find nothing

→ This is H3, and it is the failure mode we most expect. Subtract the per-frame mean before scoring. Reading Xiao et al. saved us a week here.

Open questions for the TA are listed in the roadmap document, including what counts as an acceptable interface for this project.

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Thank you

Questions and discussion

Team Claudn't · Roll numbers: [FILL IN]

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